Distilling Large Teachers into a 4.9M-Parameter ResNet for CIFAR-10

Deep Learning (Fall 2026), Project 1

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Code: `https://github.com/RoroDev2023/nyu-dl-f26-challenge`

Overview

The task was to design a residual network [1] with at most 5 million trainable parameters, trained from scratch on the competition's version of the CIFAR-10 training set [2] (50,000 images), and to maximize accuracy on a hidden 2,000-image test set. Our final model, `r3_80_2-3-3`, is a three-stage ResNet with 80/160/256 channels and 2/3/3 basic blocks (**4,880,474 parameters**). It is trained for 1,200 epochs with SGD, strong augmentation, sharpness-aware minimization (SAM), an exponential moving average (EMA) of the weights, and knowledge distillation (KD) from three much larger teachers that we trained ourselves on the same data. At test time we average the softmax of 10 shifted and flipped views.

Summary of findings.

- **Final result**: **91.1% on the Kaggle public leaderboard**; on our fixed 5,000-image validation holdout the same recipe reaches **97.92%** (600 epochs). A plain ResNet of the same size reaches 95.42%.
- Most of the gain came from the training recipe, not the architecture: augmentation + SAM + EMA gave +1.9 pt, distillation +0.4 pt, a better width/depth split +0.1–0.2 pt, test-time augmentation (TTA) +0.1 pt.
- With a fixed budget, a middle ground between depth and width won: 80/160/256 channels with 8 blocks beat a deeper (11 blocks, 64–256 channels) and a wider (6 blocks, 96–320 channels) variant.
- Many published tricks did not help on top of this recipe when tested against a matched control: stochastic depth (-18 images, p=.005), SWA, weight soups, logit-standardized KD, test-time BN adaptation, AugMix, and splitting the budget into a two-model ensemble.
- Validation accuracy (97.9%) is far above the test score (91.1%). We verified the pipeline end to end (the notebook reproduces the submitted file label for label); the remaining gap looks like a distribution shift between the training data and the hidden test images (Section 2.4).

1 Methodology

1.1 Data and evaluation protocol

We hold out a fixed, class-balanced validation set of 5,000 training images (`train_val_split(val_size=5000, seed=0)`). Every student *and every teacher* holds out the same images, so no teacher has seen a validation image. Models are compared on these 5,000 images with a *paired* test: for two models we count the images only one of them gets right and apply an exact two-sided sign test. One validation image is 0.02 pt, and differences below about 8–10 images were usually noise in repeated runs. We reuse the same validation set many times, so the p-values are exploratory. The final submission is retrained on all 50,000 images with the chosen recipe. Since that run has no holdout, we report the validation numbers of the identical 600-epoch run (A4).

1.2 Architecture

We keep the residual block exactly as specified: $\mathrm{ReLU}(S(x)+F(x))$ with F = conv→BN→ReLU→conv→BN, and look for the best split of the 5M budget over the design variables of the assignment: number of stages N, blocks per stage B_i, channels C_i, conv kernel F_i, shortcut kernel K_i, and pool size P.

Table 1: Student architectures (all 3×3 convs, 1×1 projection shortcuts, global average pool). Validation = 300 epochs, identical recipe with KD from 3 teachers, EMA weights, flip TTA.

Name	Channels C_i / blocks B_i	Params	Val (%)	Notes
r3_80_2-3-3	80, 160, 256 / 2, 3, 3	4,880,474	**97.62**	chosen
r3_96_2-2-2	96, 192, 320 / 2, 2, 2	4,902,506	97.42	widest, shallowest
r3_64_3-5-3	64, 128, 256 / 3, 5, 3	4,918,602	97.38	first design
r4_64_3-3-2-1	64, 128, 256, 320 / 3, 3, 2, 1	4,890,442	–	not finished
r3_40_8-8-8	40, 80, 160 / 8, 8, 8	4,722,610	–	not finished (slow)
WRN-40-2 (×2 ensemble)	32, 64, 128 / 6, 6, 6 (pre-act)	2×2,243,546	97.26	both together

- **Stem and pooling.** A 3×3 conv stem without max-pooling keeps the full 32×32 resolution in stage 1, which matters for 32-pixel images. The head is global average pooling (P=8, the final 8×8 map) followed by one linear layer.
- **Kernels.** F_i=3 everywhere. A 5×5 kernel costs 2.8× the parameters of a 3×3 one, and two stacked 3×3 convs already cover a 5×5 receptive field. The shortcut is the identity inside a stage, and a 1×1 conv + BN with stride 2 (K_i=1) when the resolution halves. A 3×3 shortcut would only spend budget on a path whose job is to pass information through.
- **Depth vs width.** The number of parameters of a stage grows with $B_i C_i^2$, so with a fixed budget there is a direct trade-off between depth and width. Table 1 lists the candidates (all 4.7–4.9M parameters). Three stages with 8 blocks of 80–256 channels won the 300-epoch sweep. Very deep, narrow nets (24 blocks of 40–160 channels) and a fourth stage at 4×4 resolution were slower per epoch and could not be finished within our compute; we dropped them.
- **Initialization.** He initialization; the last BN of each block starts at zero, so every block begins as the identity [3], which made the 1,200-epoch runs stable from the first epoch.

Pros and cons. Wider stages give each block more features to work with and run faster on a GPU (fewer sequential layers), but every extra block is expensive. Deeper nets reuse parameters over more nonlinear steps but are slower and, at this size, become too narrow early on (40 channels at full resolution). Spending more width at low resolution (r3_96, 320 channels in stage 3) was worse than r3_80, which moves capacity to the 16×16 stage. Our reading is that mid-resolution features matter most for 32×32 objects.

1.3 Training recipe

All models use SGD with Nesterov momentum 0.9, batch 128, peak learning rate 0.1 with 5 warmup epochs and per-step cosine decay to 0 [4], and weight decay $5 \cdot 10^{-4}$ (not on BN parameters or biases). Mixed precision (bf16) is used on A100/H100 GPUs. On top of random crop (4 px padding) + horizontal flip we add:

- **TrivialAugment** [5] (one random photometric/geometric op per image, random strength) and **Cutout** 16×16 [6].
- **mixup** (α=0.2) [7] or **CutMix** (α=1) [8], chosen at random per batch, turned off for the final 50 of 1,200 epochs (15 of 300) so the model ends on clean images.
- **SAM** [9] with ρ=0.05: each step first moves the weights in the direction that increases the loss most (inside a small ball), and then takes the gradient there. This costs a second forward/backward pass. BN running statistics from the perturbed pass are discarded.
- **EMA** of the weights with decay 0.999; we evaluate and submit the EMA weights.
- Label smoothing 0.1 [11], only when there are no teachers (with KD the soft targets already do this).

1.4 Knowledge distillation

A 5M-parameter model trained on 50k images is limited mostly by how much the labels can teach it, not by optimization. Following [12], we first train large teachers from scratch on the same 45k training images with the same recipe (400 epochs): two **WRN-28-10** [15] (36.5M parameters, val 98.06% and 98.12%) and a **PyramidNet-110** (α=270) with ShakeDrop [16, 17] (28.5M, 97.64%). The teachers are only used during training. At inference the system is the single 4.88M student.

The student loss is

$$\mathscr{L} = (1-\alpha)\,\mathrm{CE}(z_s, y) + \alpha T^2\,\mathrm{KL}(\bar{p}_T \,\|\, \mathrm{softmax}(z_s/T)), \qquad \bar{p}_T = \tfrac{1}{3}\textstyle\sum_k \mathrm{softmax}(z_k/T),$$

with α=0.9 and T=4. A key detail is *consistent teaching* [13]: the teachers see exactly the same augmented, mixed batch as the student (instead of clean images), so the soft targets describe the image the student actually sees. Together with

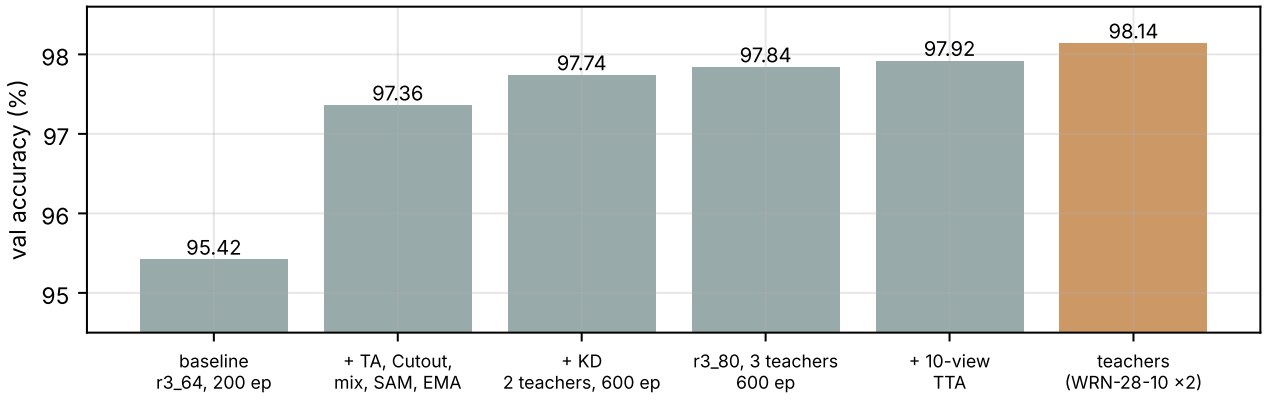


Figure 1: Validation accuracy as components are added (EMA weights; student bars use flip TTA except the 10-view TTA bar). The last bar is the best teacher pair (two WRN-28-10, 73M parameters), which bounds what distillation can reach.

heavy augmentation, this lets distillation benefit from very long schedules. Going from 600 to 1,200 epochs did not hurt on validation (97.92 vs 97.94%), and the longer schedule gave our best test score.

1.5 Test-time augmentation

We reflect-pad each test image by 4 pixels and take five 32×32 windows (centered and shifted by ±2 pixels vertically or horizontally), plus the same five windows of the mirrored image, and average the 10 softmax outputs. This adds no parameters. On validation it adds +0.08 pt over flip-only TTA. Weighting the center views more (50% or 80% of the weight) or adding diagonal shifts did not help when tuned on one half of validation and checked on the other.

1.6 Compute

Training ran on NYU HPC (A100), Modal and Vast.ai (A100/H100). With KD from three teachers and SAM, one epoch of the student takes 27 s on an A100 (18 s on an H100 with GPU-side augmentation). The final 1,200-epoch run took about 9 GPU-hours, and each WRN-28-10 teacher about 3 GPU-hours.

2 Results

2.1 Final model

Table 2: Final submission. "Val" uses the identical recipe at 600 epochs with the 5,000-image holdout (A4).

Architecture	ResNet `r3_80_2-3-3`: 3×3 stem (80 ch); stages of 2, 3, 3 basic blocks with 80, 160, 256 channels at 32^2, 16^2, 8^2 resolution; 1×1 projection shortcuts; global average pool; linear $256\to10$
Parameters	**4,880,474** trainable (budget 5,000,000; teachers not used at inference)
Training	1,200 epochs on all 50k images, seed 1; KD from 3 teachers; TA, Cutout, mixup/CutMix, SAM, EMA
Inference	EMA weights, 10-view TTA (5 shifts $\times$ flip), softmax average
Validation accuracy	**97.92%** (no TTA 97.72%, flip TTA 97.84%)
Test accuracy	**91.1%** (Kaggle public leaderboard)

2.2 Contribution of each component

Figure 1 breaks the result down. The plain baseline (`r3_64_3-5-3`, crop + flip, label smoothing, 200 epochs) reaches 95.42%. The augmentation + SAM + EMA recipe adds +1.9 pt, distillation from two WRN teachers with a 600-epoch

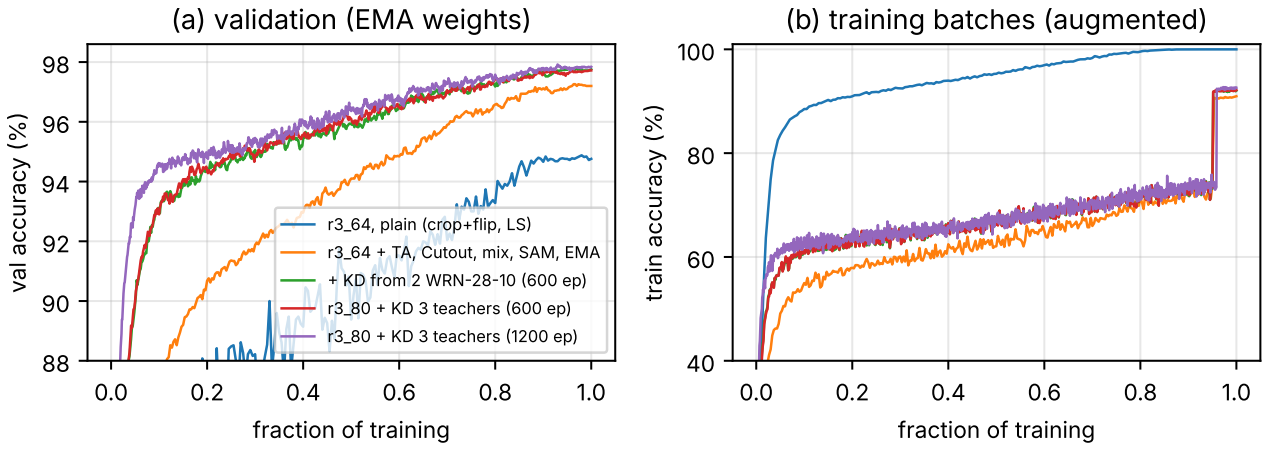


Figure 2: (a) Validation accuracy during training (x axis normalized, runs are 200–1,200 epochs). Strong augmentation makes early validation accuracy worse, but the final accuracy much better. (b) Training accuracy on augmented, mixed batches stays at 60–75% while mixing is on and only reaches about 92% after it is turned off near the end; the plain baseline memorizes its training set (100%).

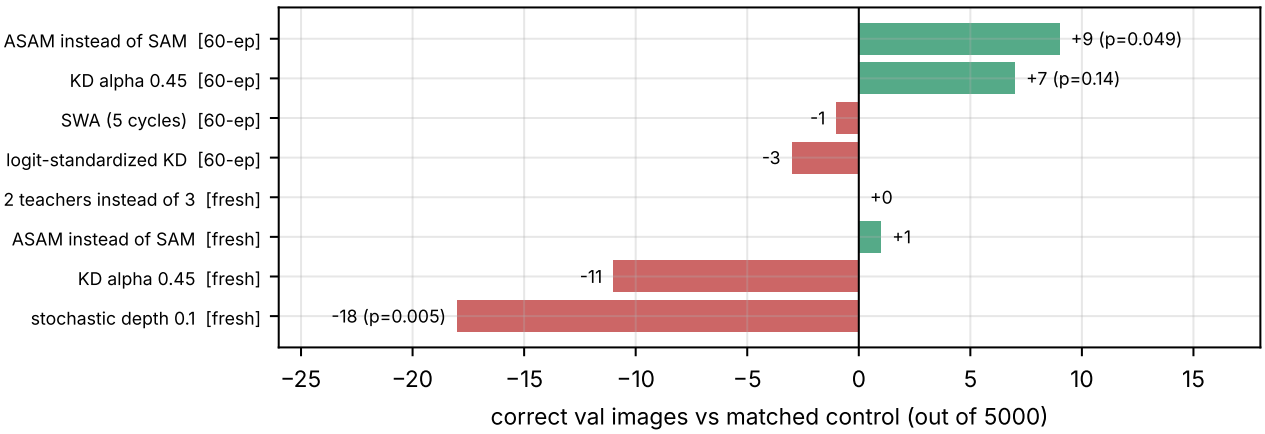


Figure 3: Paired differences against a matched control, in correctly classified validation images (out of 5,000; 5 images = 0.1 pt). "60-ep" = 60-epoch continuation of A4 (control: SAM continuation); "fresh" = 300-epoch run from scratch.

schedule +0.4 pt, the r3_80 architecture with three teachers +0.1 pt, and 10-view TTA +0.08 pt. The student ends 0.2 pt below its teachers, which have 15× more parameters. Figure 2 shows that the baseline fits its training set perfectly while the regularized models never do. This is the main reason long schedules keep helping instead of overfitting.

2.3 What did not help

We judged every idea against a matched control (same start point or same seed and recipe, one change) on the same validation images. Figure 3 shows the controlled comparisons. Table 3 lists the rest.

2.4 Validation vs. test gap

Our validation accuracy (97.9%) is about 7 pt above the public test score (91.1%). We checked the obvious causes: no overlap between training and validation images; the exact submission code gives 97.82% on validation; the notebook regenerates the submitted CSV with 0 differing labels; per-class validation accuracy is unremarkable. What we did measure is that the test images differ from the training images: they are brighter (mean RGB 0.513/0.498/0.461 vs 0.491/0.482/0.447), have higher contrast, are about 7% sharper and less saturated. The student is less confident on test (0.85 vs 0.90 mean max-

Table 3: Other experiments. Accuracy on validation (10-view TTA, EMA) unless marked as public leaderboard (LB).

Experiment	Result
ASAM [10] instead of SAM	+9 images as a 60-epoch continuation (p=.049), confirmed weakly from a second start point (+1/+4), but no continuation beat its *untouched* start model. From scratch (300 ep): +1 image. On the test set, ASAM finish of F2s1: LB 0.908 vs 0.911.
Stochastic depth 0.1 [18]	-18 images (p=.005) and worse calibration: on top of KD + SAM + mixing, extra regularization hurts.
SWA [19]	6 cyclic 10-epoch cycles, averaged, BN recomputed: -1 image vs control, -6 vs start.
Weight soup [20]	Average of a model and its fine-tuned version: $-1/-3$ images; on test 0.908.
Logit-standardized KD [14]	Same accuracy as ordinary KD (97.62 vs 97.62, 300 ep), much better calibration (val NLL 0.09 vs 0.15).
KD α=0.45; 2 vs 3 teachers	α=0.45: -11 images from scratch. Dropping the PyramidNet teacher: ± 0.
Two WRN-40-2 (ensemble)	2.24M each, 97.10/97.06 alone, 97.26 averaged vs 97.62 for one 4.88M ResNet: one model is better than two half-size models.
AugMix + JSD [21]	80-ep finish: +0.6 pt on blurred validation images but -0.6 pt on clean images.
Test-time BN adaptation	Re-estimating BN statistics on the test images: val 97.34, LB 0.884 (worse). BN recalibration on clean train images: -0.6 to -0.8 pt.
Class-balanced prediction	Sinkhorn balancing [22] of the test predictions: LB 0.898 vs 0.899.
More epochs	600→1,200 epochs: 97.92→97.94 (+1 image). A random seed changes 61/2000 test labels and up to 1.3 pt on the LB.

probability) and agrees less with its teachers (96.9% vs 99.2%). However, simulating these pixel statistics on validation costs only 0.1–0.3 pt. Our hypothesis, which we could not test without test labels, is a content-level shift like CIFAR-10.1 [23], where re-collected test images cost CIFAR-10 models 4–10 pt. A consequence is that public LB differences of about ± 0.5 pt between runs were mostly noise (2,000 images, seed variance), so we made decisions on the paired validation tests and used the leaderboard only as a final check.

3 Lessons learned

- **The recipe matters more than the architecture.** Among sensible 5M ResNets the spread was 0.25 pt. Augmentation, SAM, EMA and distillation together were worth about 2.3 pt.
- **Distillation needs strong, matching teachers.** Teachers trained with the same augmentation, fed the student's own augmented batch, made long schedules (600–1,200 epochs) useful for a small model.
- **Do not stack regularizers blindly.** Once the model is strongly regularized (it never fits its training set), stochastic depth, AugMix and lower KD weight all hurt.
- **Test against a matched control.** Many ideas looked good against an earlier, different baseline and disappeared in a paired comparison. Late "finishing" runs from a converged checkpoint never improved on the checkpoint itself.
- **One budget, one model.** Two half-size models averaged lost 0.36 pt to one full-size model.
- **Be careful with a small public test set.** With 2,000 test images and a shifted distribution, seed noise on the leaderboard is as large as most real improvements.

Code

The repository contains the training code (`train.py`, all tricks as flags), models (`models.py`), KD/SAM/EMA/mixing (`tricks.py`), the single TTA implementation (`tta.py`), evaluation scripts, the per-run logs and arguments (`runs/`), a full experiment log (`EXPERIMENTS.md`), and `report.ipynb`, which prints the parameter count, reproduces the submission and the validation accuracy, and makes every figure in this report. Parts of our setup (no max-pool stem for 32×32 inputs, standard CIFAR training pipeline) follow common practice as in [24]. Teacher architectures follow the original papers.

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